

Topic 6E: Alcohols

37. know that alcohols can be classified as primary, secondary or tertiary

38. understand the reactions of alcohols with:

- i oxygen in air (combustion)
- ii halogenating agents:
 - PCl_5 to produce chloroalkanes
 - 50% concentrated sulfuric acid and potassium bromide to produce bromoalkanes
 - red phosphorus and iodine to produce iodoalkanes
- iii potassium dichromate(VI) in dilute sulfuric acid to oxidise primary alcohols to aldehydes (including a test for the aldehyde using Benedict's/Fehling's solution) and carboxylic acids, and secondary alcohols to ketones

In equations, the oxidising agent can be represented as [O].

- iv concentrated phosphoric acid to form alkenes by elimination

Descriptions of the mechanisms of these reactions are not expected.

39. understand the following techniques used in the preparation and purification of a liquid organic compound:

- i heating under reflux
- ii extraction with a solvent in a separating funnel
- iii distillation
- iv drying with an anhydrous salt
- v boiling temperature determination

CORE PRACTICAL 5: The oxidation of ethanol**CORE PRACTICAL 6: Chlorination of 2-methylpropan-2-ol using concentrated hydrochloric acid**

Topic 7: Modern Analytical Techniques I

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Hands-on practical work is limited in this topic, although many universities allow students to visit and learn about instrumentation first hand.

Mathematical skills that could be developed in this topic include analysing fragmentation patterns in mass spectra.

Within this topic, students can consider how different instrumental methods can provide evidence for analysis. They can see how accurate and sensitive methods of analysis can be applied to the study of chemical changes, but also to detect drugs such as in blood or urine testing in sport.